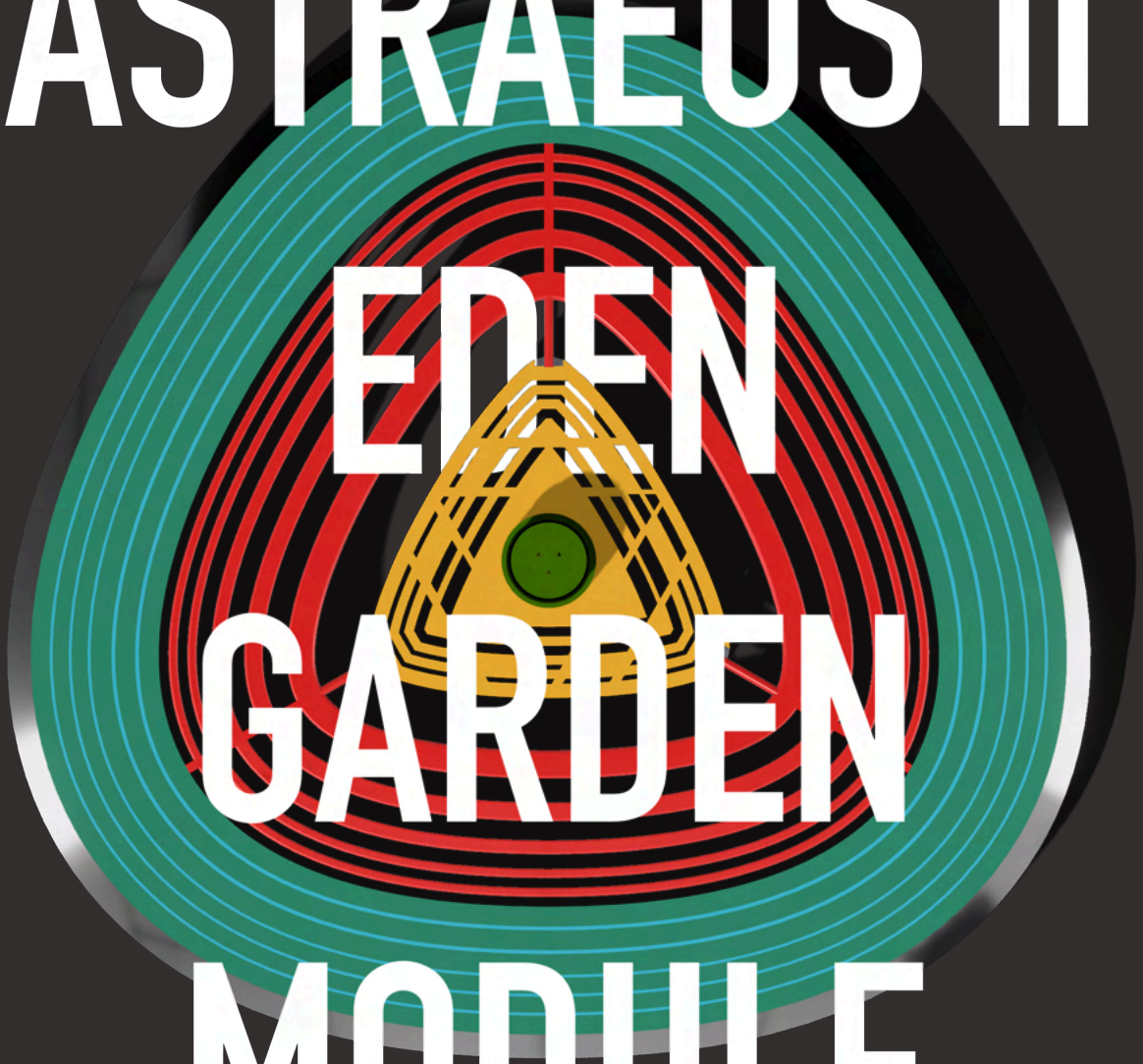
ASTRAEUS II



EDEN GARDEN MODULE

A comprehensive report prepared by: *Aaron Kumar, Aadit Gupta, Adwaith Shaji, Armaan Singh, Devansh Pavithran, Medhansh Kapoor, Parmeet Abbat, Sahib Sohi, Sanil Manandhar, Shivam Patel, Soham Soni, Soumya Soni*

EXECUTIVE SUMMARY

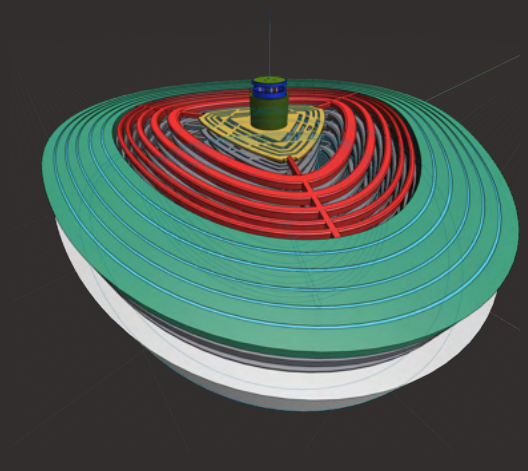
The ASTRAEUS II: EDEN GARDEN MODULE stands out for its innovative use of ***magnetorheological fluid*** and its near ***100% space-efficient*** agricultural module. The fluid, synthesized using green tea extract and ferric chloride, solidifies under magnetic fields, enabling precise, non-mechanical soil tilling—a first for space farming. The agricultural module maximizes space utilization, integrating hydroponic, aeroponic, and soil-based systems into a closed-loop design. This near-perfect efficiency ensures sustainable, scalable food production while reducing resource waste.

Together, these breakthroughs redefine the possibilities for sustainable living in space. The module has a number of other ground breaking features such as an optimized frequency system to increase plant growth, an automated pollination vehicle, and an advanced hyd roponic system with LED optimization.

The EDEN GARDEN MODULE is replicable; in order to make one complete sector for farming, the module stacks upon itself 16 times. In a single sector 24 m diameter x 54.4 m height, there lies 1.661 acres of farming space. The ASTRAEUS II settlement will require approximately 262 of these modules for a combined 435.182 acres (18,956,527.92 square ft) of farming space. This will comfortably support—with no consideration to living space—296,196 inhabitants with no preservation of food or 592,391 with preservation of food inhabitants on the ASTRAEUS II settlement.

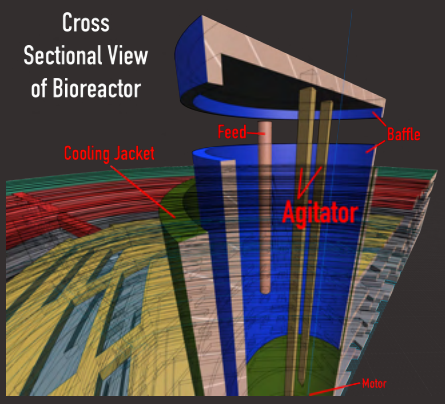
Generative AI was not used in this entry.

Agriculture Module Design

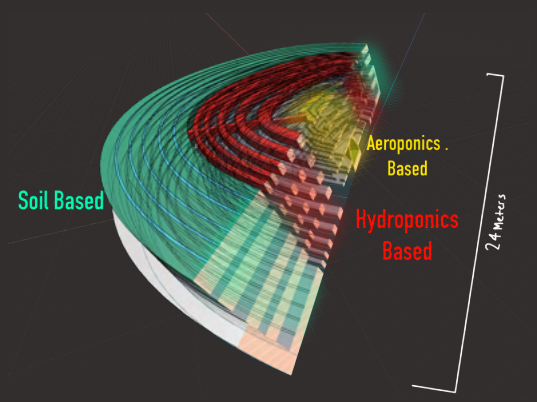


This integrated agricultural system includes a bioreactor, aeroponic, hydroponic, and soil-based systems, all interconnected through a water distribution network to optimize plant growth efficiently. This module is replicable and attaches easily to the main ASTRAEUS II design. Below are the components:

- Bioreactor (Green Section)
    - Centrally located for nutrient distribution. Features include a motor, agitator, cooling jacket, and baffles for efficient microbial processing of organic waste into nutrients.



- Aeroponic System (Yellow Section)
    - Located next to the bioreactor. Uses misting to deliver nutrients and water to plant roots with minimal water usage.
- Hydroponic System (Red Section)
    - Positioned adjacent to the aeroponic system. Submerges plant roots in nutrient-rich water, with recirculation to prevent wastage.
- Soil-Based System (Blue Section)
    - Located at the perimeter. Uses magnetic fluid for automated tilling, aerating soil 1efficiently through magnet activation.



- System Integration
    - A closed-loop piping network connects all components, circulating water and nutrients for sustainable operation.

- The cross-sectional images show the green bioreactor in the center, surrounded by the yellow aeroponic, red hydroponic, and blue soil-based systems.

Tilling — Using Magnetorheological Fluid

Concepts Behind Tilling

Tilling soil involves turning over and breaking up the surface soil and is vital to cultivating a successful farm. Tilling soil helps mix fertiliser into the soil, encourages root growth of plants, and allows water/nutrients to go deeper into the soil. Tilling usually involves 2 rounds, the first, is to break the tough, outer layer of the soil, and remove weeds, while the second round makes the soil finer. Tilling soil is also helpful for getting rid of pests. A critical aspect of tilling involves not over-tilling the soil. Excessive tillage can increase the chance of soil erosion, damaging the soil, and making it unusable.

Breaking The Boundaries Of Agriculture

Most systems on earth use some mechanism of blades that disturb the soil, as it gives the best combination of efficiency, disturbing the dirt, and removing weeds. On a space settlement, however, we do not need to worry about weeds, and we do not necessarily have to use a system involving blades. Instead, we suggest using magnetism to our advantage to disturb the soil efficiently. By using a fluid, known as magnetorheological fluid, and its properties, we can efficiently disturb the soil. This fluid solidifies when exposed to a magnetic field, therefore, if a powerful magnet is placed underneath the farming structure, the fluid will poke into the soil. By moving containers holding the soil horizontally, the soil will effectively be distributed.

Creating The Fluid

The first step is to combine a green tea extract with ferric chloride. According to an article by NIH, using Green Tea extract and combining it with

Finding The Mass of Ferric Chloride

125mL of $FeCl_3$ at 0.1M.

Molar Mass $FeCl_3 = 55.85 + 35.45 \cdot 3 = 162.2g$

$C = \frac{n}{V}$ $\quad n = \frac{m}{M}$

$0.1 = \frac{n}{0.125}$ $\quad 0.0125 = \frac{m}{162.2}$

$n = 0.0125mols$ $\quad m = 2.03g$ of Ferric Chloride

$\therefore$ 2.03g of Ferric chloride is required

ferric chloride would lead to the creation of magnetic iron nanoparticles. We had a total of 250ml of green tea extract and needed 125mL of Ferric Chloride at a concentration of 0.1 M. Using Stoichiometry, we calculated the mass of ferric chloride needed to dilute. This amount was determined to be 2.03 g. Therefore the procedure was to create 250mL of green tea extract and dilute 2.03 g of ferric chloride into 125mL of distilled water.

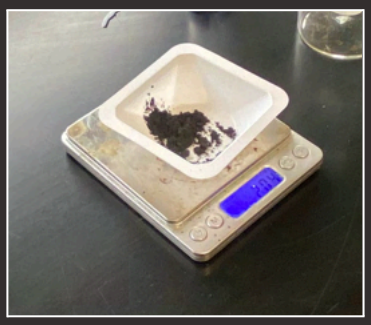
The Timeline Of Creation:

Note: If any pictures are unclear or too small please zoom into the document.

Step 1: The Green Tea Extract

250 mL of Green Tea Extract was created by steeping tea in distilled water.

Step 2: The Ferric Chloride



2.04 g of Ferric Chloride was measured using a scale (as close as possible to 2.03g)

Step 3: The Ferric Chloride Solution

The Ferric Chloride was diluted with 125mL of distilled water to make a Ferric Chloride solution (orange solution) to be combined with the extract (reddish solution).

Step 4: Combining The Two Solutions

The 125mL Ferric Chloride solution was combined with the 250mL of Green Tea Extract.

Step 5: The Combined Solution

Using a stirring rod, the two solutions are mixed until the resultant solution is dark black.

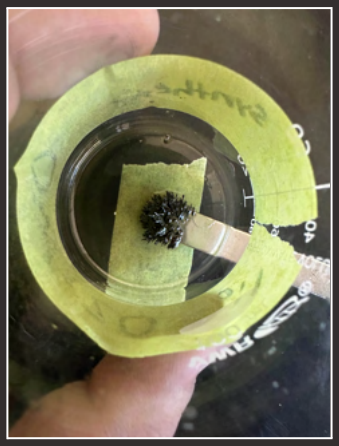
Step 6: Collecting The Iron Particles

A strong magnet was placed inside a beaker with the solution (a smaller beaker for convenience).

Step 7: Extract and Repeat

The iron particles were placed into a white dish, and more were collected by placing the magnet inside the black solution. As you can see, with a magnet at the bottom of the dish, the particles spiked up.

Step 8: Motor Oil



A few drops of motor oil were added to the iron particles to create a fluid-like substance. As you can see, when the magnet is below the beaker the substance spikes up.

Significance Of Our Findings

The experiment above is a small showcase of these particles which will be applied to a much larger scale. Instead of having less than a gram's worth of yield, on the spaceship, we would have kilograms worth. This magnetic “fluid” would be placed inside malleable containers hidden within the top ledge of the farm. When the soil needs to be tilled, the malleable containers will be extended above the top layer of soil, and a strong magnet at the bottom of the farm will be activated. This will

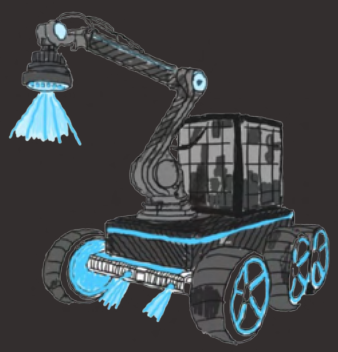
cause the fluid to become a spiky solid, as seen in step 9 of the timeline. Since the containers are malleable they would take the shape of these spikes, and then the containers can be shifted left and right to disturb the soil. Once the tilling process is done, the magnet can be deactivated and then the fluid will no longer be spiky. This can mean the containers can be retracted back into the side of the farms, making the process time and space-efficient compared to manually tilling using large machines.

The Automated Pollination Vehicle

The Automated Pollination Vehicle is designed for environments, such as indoor farms and space habitats, that lack natural pollinators. This compact, high-technology vehicle uses advanced sensors and a pollen dispersal system, along with AI, for precise and efficient pollination; it then moves independently and adjusts schedules based on real-time environmental feedback.

Vehicle Description

The APV has a rugged yet lightweight carbon fiber frame with main components: sensors, an AI processor, battery, pneumatic pump, and pollen reservoir—all protected by tempered glass. An AI processor, such as the NVIDIA Jetson Xavier NX, processes the sensor data in order to map the environment, identify plants, and avoid obstacles in the path. LIDAR creates a 3D map while cameras detect crop rows, and environmental sensors optimize pollen dispensing for better efficiency. The four brushless DC motors make the robot maintain high-precision movement along the plant rows.



Rough Sketch of the APV

The pneumatic pump, regulated by AI and a solenoid valve, makes direct pollen delivery to flowers. Further, the clog-resistant reservoir and retractable nozzle ensure continuous and precise application of pollen. With a large-capacity battery, the APV has efficient power management with

automatic docking in view of charging. The 5G module and connected dashboard enable real-time monitoring and adjustment of the pollination schedule.

Vehicle Operation

The APV is autonomous, initiating a self-test to ensure its systems are running. It then charts a map of the environment and moves autonomously along rows of crops, detecting obstacles and navigating efficiently to the right location. Having detected a flower, it precisely orients its nozzle, releases pollen, and checks the environmental conditions to maximise pollen adhesion. After having completed the pollination mission, the APV autonomously returns to the charging station, uploads its data, and places itself in the starting position for the next pollination cycle.

LED Light Optimization

The wavelength and duration of the exposure to light are really important to ensure maximum growth by the crops. During vegetative growth, which generally lasts 4–8 weeks, the most effective light is blue, in the 400–500 nm range. It improves the efficiency of photosynthesis, facilitating the exchange of gases by opening the stomata, which favors the growth of strong stems and leaves. Once crops enter the flowering or fruiting stage, then the 600–700 nm range of red light becomes more effective. Red light signals the end of vegetative growth and increases energy production for photosynthesis; it also activates phytochromes, which control blooming, maturation, and fruit quality. Light duration is also an important factor. Long-day plants require more than 12 hours of light, while short-day plants need less than 12 hours. Day-neutral plants will grow as long as some minimum light quantity is met. Specifically, crops such as tomatoes and beans do best with 12–16 hours of light per day, while deviation from this range may impact flowering and fruiting. For a space settlement, blue light is used for the first 4–8 weeks to promote vegetative growth; red light should be used for 4–8 weeks if flowering crops are desired, or 2–10 weeks if fruiting crops are desired. This will also help to keep the day length constant at 14 hours per day.

LED Efficiency

Key areas in ensuring crop growth in hydroponics include speed and LED optimization, but just as important is the longevity and efficiency of the system. Hydroponic systems re-use nutrient-enriched water to increase yields by 20-50% over soil-based growing while conserving over 90% of water. Canadian ag-tech company Zipgrow has managed to lower water loss to only 1.5%, but for space applications, this needs to be minimized to less than 1% for efficiency and sustainability. Evapotranspiration can be difficult to control, but temperature-range optimization may help to minimize that water loss, making hydroponics more efficient for resource-constrained environments like space.

Hydroponics

A method of growing plants that doesn't involve the use of soil. This is incredibly convenient for space travel, because it uses less resources and space to grow more produce. Prepackaged meals can be heavy, especially for multi-year voyages. It's a lot more convenient to start a hydroponic farm and grow fresh produce.

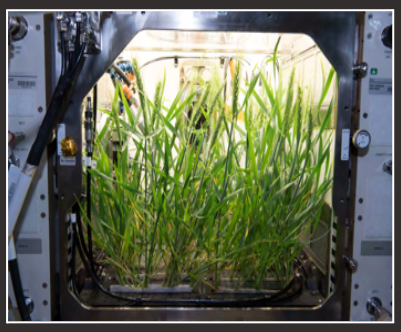
Veggie

The Vegetable Production System, or Veggie, is a hydroponic system on the International Space Station (ISS) that provides salad crops for the crew. It uses a syringe to water plants in containers called plant pillows. Veggie also uses the cabin's temperature and carbon dioxide to promote growth.

Advanced Plant Habitat (APH)

A hydroponic growth chamber developed as a better alternative to Veggie, that uses LED lights, a porous clay substrate, and controlled-release fertiliser to deliver water, nutrients, and oxygen

to the plants. The APH is mostly automated and covered in cameras (180 sensors). The automation



controls water recovery and distribution, atmosphere content, moisture levels, and temperature. It has more LED lights than Veggie, with red, green, and blue lights, but also white, far red, and even infrared to allow for nighttime imaging. *APH had its first test run on the space station in Spring 2018 using Arabidopsis thaliana (the "white mouse of the plant research world") and dwarf wheat.*

Hydrogels

A substrate that can provide water and nutrients for plants in space. Hydrogels can be incorporated with on-site regolith to create Earth-like soil.

LED Lights

The ISS uses LED lights to help guide plants out of the growth media. Plants absorb more red and blue light, and reflect green, so the LEDs in the ISS' farms grow a bright magenta.

Plant Frequency Project

Music can affect emotions; plants, even without ears or emotions, are no exception. Their cells interact with sound frequencies, which encourage plant growth by enhancing the circulation of nutrients, root strength, and vitality of leaves. Research indicates that plants may have preferences in what they like to hear. Classical or jazz music encourages growth, but heavy metal may stress plants. This has led to the development of the PAFT system: Plant Acoustic Frequency System, which uses tailored frequencies in order to enhance processes like photosynthesis, hence giving stronger plants and better yields, particularly in vegetables.

Different Frequencies

125Hz: Enhances root growth by stimulating mechanical resonance, aiding water and nutrient absorption (e.g., studies by Dr. Stefano Mancuso in plant neurobiology).

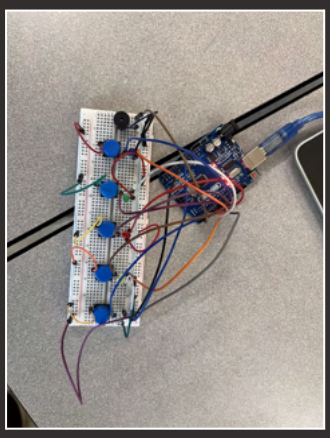
250Hz: Stimulates auxin distribution, promoting cell elongation and stem growth.

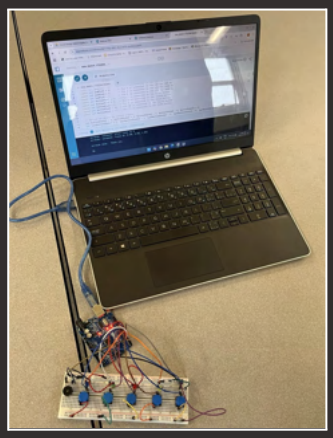
500Hz: Boosts cellular activity and enzyme production, supporting overall growth (e.g., 2010 *Journal of Experimental Botany* study).

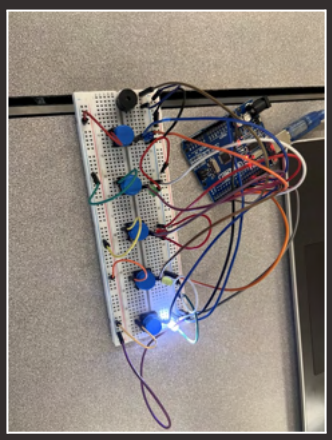
1,000Hz: Activates stomatal opening, increasing CO2 uptake and photosynthesis (e.g., research by the Agricultural University of China).

8,000Hz: Triggers stress responses that enhance flowering and fruiting, mimicking environmental cues like wind or insect activity.

Note: *The Arduino Circuit created is a small scaled version of how different frequencies will be created, operated and used to amplify plant growth based on the research above. (Appendix C)*







Food Science (Meal Plan - Appendix A)

Breakfast: Quinoa Bowl with Berries and Herbal Tea (Appendix A)	**Lunch: Lentil, Kale, and Turmeric Stew with Flatbread**	**Dinner: Mediterranean Stuffed Eggplant with Lentil-Tahini Sauce**
Quinoa Bowl (Minimally processed, high in protein, and low in glycemic index) • Cooked quinoa (space-grown, minimally processed): 50g • Almonds (soaked overnight for improved digestion): 10g	Stew (High in protein and fiber for insulin stabilization) with Turmeric (Proven to reduce inflammation) • Cooked lentils (rich in protein and fiber): 100g • Hydroponically grown kale: 50g • Space-grown carrots: 50g	Stuffed Eggplants (Provide high-quality plant protein and complex carbs with a low glycemic index.): • Medium eggplants (aeroponic): 2 • Cooked chickpeas: 75g • Space-grown onions: 20g • Hydroponic spinach: 50g

● Chia seeds: 10g ● Freeze-dried blueberries: 20g ● Unsweetened almond milk: 200ml ● Natural cinnamon: 1 tsp Herbal Tea (Supports digestion and hydration) ● Mint leaves (aeroponic): 5g ● Boiled recycled water: 200ml	● Tomatoes (pureed): 100g ● Olive oil: 1 tbsp ● Garlic: 1 tsp ● Turmeric (anti-inflammatory): 1 tsp ● Recycled water: 200ml Flatbread (Minimal processing and sustained energy) ● Whole grain flour (aeroponic wheat): 50g ● Rehydrated yeast: 1 tsp ● Recycled Water: 50ml	● Olive oil: 1 tbsp ● Paprika: 1 tsp ● Cumin: 1 tsp Lentil-Tahini Sauce (Rich in healthy fats, supporting anti-inflammatory benefits): ● Cooked lentils (pureed): 50g ● Tahini (space-processed from sesame seeds): 10g ● Lemon juice: 1 tsp ● Garlic: 1 clove ● Olive oil: 1 tsp

Nutritional Coverage

Nutrient	Daily Requirement	Provided by Plan	Key Sources
Calories	~2000 kcal	~2000 kcal	All meals evenly distribute energy
Protein	~75g	~78g	Quinoa, lentils, chickpeas, tahini
Carbohydrates	~225g (low GI)	~225g (low GI)	Quinoa, lentils, whole grain
Fiber	~30g	~37g	Lentils, quinoa, vegetables, nuts
Healthy Fats	~70g	~75g	Almonds, tahini, olive oil
Vitamins & Minerals	100% RDAs	Met	Kale, spinach, carrots, eggplants

This 1-day sample diet tackles the major health issues astronauts have in space: muscle atrophy, bone density loss, and inflammation. High-quality protein from quinoa, lentils, and chickpeas will help to repair muscles, while tahini, leafy greens, and legumes will provide enough calcium and magnesium to support bone health. High-fiber foods like nuts and vegetables further help with digestion in microgravity. Other anti-inflammatory compounds of turmeric, olive oil, and spinach also help to modulate chronic inflammation. It contains sustainable, space-growable crops such as kale and eggplants, which have minimal processing to retain their nutrients. A diverse meal supports both physical health and psychological well-being to keep metabolic balance.

Appendix A

Images for Food Plan

Quinoa Bowl

Lentil, Kale, and Turmeric Stew

Eggplant Curry

Note: *All images in this appendix were sourced from their Canadian Food Guide Recipe Page (cited)*

Appendix B

Links to Recipes

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Appendix C

Video of Plant Frequency Project

https://youtube.com/shorts/lyNdNiikIS0

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